

Mid-Project Update Report

Multivariate Forecasting and Predictability Analysis of Cryptocurrency Market Dynamics

2026 Capstone Group 27 A

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1.0 Business Problem

1.1 Problem Context

The Data Science Institute is the client organisation for this project. It focuses on applying data science, machine learning, and advanced analytics techniques to better understand complex datasets and generate meaningful insights. The institute works with researchers, students, and industry partners to explore how data-driven approaches can be used to address real-world analytical challenges. In this project, the Data Science Institute has engaged our team to analyse cryptocurrency market data and investigate the predictability of key market variables such as price, return, volatility, and trading volume using data mining and machine learning techniques.

Cryptocurrency markets are well known for their extremely high volatility, rapid regime changes, and strong uncertainty. Due to these characteristics, predicting cryptocurrency market behaviour is considered more difficult than predicting traditional financial market behaviour. Unlike traditional financial datasets where observations are often independent, cryptocurrency market data forms a complex time-series dataset with strong temporal dependencies. Market variables such as price, return, volatility, and trading volume are influenced by past market behaviour and can exhibit nonlinear dependencies. In addition, cryptocurrency markets often display non-stationary behaviour, and this makes statistical properties such as mean and variance change over time.

From a data mining perspective, we can formulate the problem as a multivariate time-series forecasting task. The objective is to use historical market data to predict future values of several key market variables. The client's goal is not to build a trading system, but to gain evidence-based insight into how predictable these market dynamics are. The findings will contribute to the Institute's broader research agenda on machine learning applications in complex, real-world environments.

1.2 Project Objective

This project investigates the extent to which data mining models can predict key time-series variables in a non-stationary environment, using the cryptocurrency market as a testbed. Our project is centred around forecasting four key market variables: price, return, volatility, and trading volume. These variables capture important aspects of market behaviour and allow the study to assess predictive performance across different dimensions of the cryptocurrency market.

To address this problem, the project evaluates a baseline regression model (linear regression), a Random Forest model (random forest regressor), and a neural network model (LSTM). Using historical data from Bitcoin and Ethereum sourced from Binance's public API, the project compares model performance across all four target variables. Performance is measured using RMSE, MAE, MAPE, and Mean Directional Accuracy (MDA). The final outputs will include this written report, a GitHub code repository, and a presentation to the client.

2.0 Data Exploration

2.1 Data Source

The dataset used in this project was obtained from Binance's public market data API. This source was selected because it provides historical cryptocurrency trading data for our targeted cryptocurrencies, Bitcoin and Ethereum. The dataset covers the period from 2021-01-01 to 2025-12-31 and is recorded at daily intervals.

Historical market data was collected using a Python script that accessed the API to retrieve daily candlestick (kline) data for the BTCUSDT and ETHUSDT trading pairs. Specifically, the script automatically downloaded the data for the specified time range and extracted key trading variables from the API response. The raw data was then processed and organised into an XLSX dataset with separate sheets for each cryptocurrency.

2.2 Data Description

The raw dataset consists of 10 original market variables and 1826 rows for both BTCUSDT and ETHUSDT. The 10 variables are date, open, high, low, close, trading volume, quote asset volume, number of trades, taker buy base asset volume, and taker buy quote asset volume.

After feature engineering, the number of variables has increased from 10 to 30 per asset by adding 20 derived variables. These include daily return, log return, rolling volatility over 7, 14, and 30 days, simple moving averages over 7, 14, and 30 days, exponential moving averages over 7, 14, and 30 days, RSI over 7, 14, and 30 days, momentum over 7, 14, and 30 days, and volume moving averages over 7, 14, and 30 days.

After cleaning, the final dataset retains 1796 observations per asset after removing rows affected by indicator-related missing values.

Feature	Data Type	Description
date	Date	Trading date of the observation.
open	Float	Opening price at the beginning of the trading day.
high	Float	Highest price during the trading day.
low	Float	Lowest price during the trading day.
close	Float	Closing price at the end of the trading day.
volume	Float	Total trading volume during the trading day.

quote_asset_volume	Float	Total traded value in the quote asset during the trading day.
number_of_trades	Integer	Total number of executed trades during the trading day.
taker_buy_base_asset_volume	Float	Volume of the base asset bought by takers during the trading day.
taker_buy_quote_asset_volume	Float	Value of the quote asset spent by takers buying during the trading day.
daily_return	Float	Percentage change in closing price compared with the previous day.
log_return	Float	Logarithmic return of the closing price relative to the previous day.
volatility_7	Float	Rolling standard deviation of returns over the past 7 days.
volatility_14	Float	Rolling standard deviation of returns over the past 14 days.
volatility_30	Float	Rolling standard deviation of returns over the past 30 days.
MA_7	Float	7-day simple moving average of the closing price.
MA_14	Float	14-day simple moving average of the closing price.
MA_30	Float	30-day simple moving average of the closing price.
EMA_7	Float	7-day exponential moving average of the closing price.
EMA_14	Float	14-day exponential moving average of the closing price.
EMA_30	Float	30-day exponential moving average of the closing price.
RSI_7	Float	7-day Relative Strength Index.
RSI_14	Float	14-day Relative Strength Index.

RSI_30	Float	30-day Relative Strength Index.
momentum_7	Float	Price momentum calculated over the past 7 days.
momentum_14	Float	Price momentum calculated over the past 14 days.
momentum_30	Float	Price momentum calculated over the past 30 days.
volume_MA_7	Float	7-day moving average of trading volume.
volume_MA_14	Float	14-day moving average of trading volume.
volume_MA_30	Float	30-day moving average of trading volume.

Above is the data dictionary for both BTCUSDT and ETHUSDT datasets. All of these variables will be used for finding our project's target outcomes: price, return, volatility, and trading volume.

2.3 Data Preprocessing

```

=== BTC INFO ===
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1826 entries, 0 to 1825
Data columns (total 10 columns):
#   Column                                Non-Null Count  Dtype
---  ---                                -
0   date                                  1826 non-null   datetime64[ns]
1   open                                  1826 non-null   float64
2   high                                  1826 non-null   float64
3   low                                   1826 non-null   float64
4   close                                 1826 non-null   float64
5   volume                               1826 non-null   float64
6   quote_asset_volume                   1826 non-null   float64
7   number_of_trades                     1826 non-null   int64
8   taker_buy_base_asset_volume          1826 non-null   float64
9   taker_buy_quote_asset_volume         1826 non-null   float64
dtypes: datetime64[ns](1), float64(8), int64(1)

=== ETH INFO ===
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1826 entries, 0 to 1825
Data columns (total 10 columns):
#   Column                                Non-Null Count  Dtype
---  ---                                -
0   date                                  1826 non-null   datetime64[ns]
1   open                                  1826 non-null   float64
2   high                                  1826 non-null   float64
3   low                                   1826 non-null   float64
4   close                                 1826 non-null   float64
5   volume                               1826 non-null   float64
6   quote_asset_volume                   1826 non-null   float64
7   number_of_trades                     1826 non-null   int64
8   taker_buy_base_asset_volume          1826 non-null   float64
9   taker_buy_quote_asset_volume         1826 non-null   float64
dtypes: datetime64[ns](1), float64(8), int64(1)

```

After we obtained the datasets, we checked the data quality step by step. Firstly, we examined the overview of the datasets. From the figure, we can see that both datasets have the same number of rows and columns. The data types for each column are correct, so no need to do anything about them.

```

=== BTC MISSING VALUES ===
date                0
open                0
high                0
low                 0
close               0
volume              0
quote_asset_volume  0
number_of_trades    0
taker_buy_base_asset_volume  0
taker_buy_quote_asset_volume  0
dtype: int64

=== ETH MISSING VALUES ===
date                0
open                0
high                0
low                 0
close               0
volume              0
quote_asset_volume  0
number_of_trades    0
taker_buy_base_asset_volume  0
taker_buy_quote_asset_volume  0
dtype: int64

=== BTC DUPLICATES ===
Duplicate rows: 0
Duplicate dates: 0

=== ETH DUPLICATES ===
Duplicate rows: 0
Duplicate dates: 0

```

Next, we checked for missing entries and duplicate records, especially the date column. Refer to the figure above, there are no missing values or duplicates found in both datasets.

```

print("BTC:", btc["date"].is_monotonic_increasing)
print("ETH:", eth["date"].is_monotonic_increasing)

BTC: True
ETH: True

```

Also, the date column was checked and confirmed to be in chronological order for both BTC and ETH datasets.

```

=== BTC NEW INFO ===
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1826 entries, 0 to 1825
Data columns (total 30 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   date                                1826 non-null  datetime64[ns]
1   open                                1826 non-null  float64
2   high                                1826 non-null  float64
3   low                                 1826 non-null  float64
4   close                               1826 non-null  float64
5   volume                              1826 non-null  float64
6   quote_asset_volume                  1826 non-null  float64
7   number_of_trades                    1826 non-null  int64
8   taker_buy_base_asset_volume          1826 non-null  float64
9   taker_buy_quote_asset_volume          1826 non-null  float64
10  daily_return                         1825 non-null  float64
11  log_return                          1825 non-null  float64
12  volatility_7                         1819 non-null  float64
13  volatility_14                        1812 non-null  float64
14  volatility_30                        1796 non-null  float64
15  MA_7                                 1820 non-null  float64
16  MA_14                                1813 non-null  float64
17  MA_30                                1797 non-null  float64
18  EMA_7                                 1826 non-null  float64
19  EMA_14                                1826 non-null  float64
20  EMA_30                                1826 non-null  float64
21  momentum_7                          1819 non-null  float64
22  momentum_14                          1812 non-null  float64
23  momentum_30                          1796 non-null  float64
24  volume_MA_7                          1820 non-null  float64
25  volume_MA_14                         1813 non-null  float64
26  volume_MA_30                         1797 non-null  float64
27  RSI_7                               1819 non-null  float64
28  RSI_14                              1812 non-null  float64
29  RSI_30                              1796 non-null  float64
dtypes: datetime64[ns](1), float64(28), int64(1)

=== ETH NEW INFO ===
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1826 entries, 0 to 1825
Data columns (total 30 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   date                                1826 non-null  datetime64[ns]
1   open                                1826 non-null  float64
2   high                                1826 non-null  float64
3   low                                 1826 non-null  float64
4   close                               1826 non-null  float64
5   volume                              1826 non-null  float64
6   quote_asset_volume                  1826 non-null  float64
7   number_of_trades                    1826 non-null  int64
8   taker_buy_base_asset_volume          1826 non-null  float64
9   taker_buy_quote_asset_volume          1826 non-null  float64
10  daily_return                         1825 non-null  float64
11  log_return                          1825 non-null  float64
12  volatility_7                         1819 non-null  float64
13  volatility_14                        1812 non-null  float64
14  volatility_30                        1796 non-null  float64
15  MA_7                                 1820 non-null  float64
16  MA_14                                1813 non-null  float64
17  MA_30                                1797 non-null  float64
18  EMA_7                                 1826 non-null  float64
19  EMA_14                                1826 non-null  float64
20  EMA_30                                1826 non-null  float64
21  momentum_7                          1819 non-null  float64
22  momentum_14                          1812 non-null  float64
23  momentum_30                          1796 non-null  float64
24  volume_MA_7                          1820 non-null  float64
25  volume_MA_14                         1813 non-null  float64
26  volume_MA_30                         1797 non-null  float64
27  RSI_7                               1819 non-null  float64
28  RSI_14                              1812 non-null  float64
29  RSI_30                              1796 non-null  float64
dtypes: datetime64[ns](1), float64(28), int64(1)

```

After feature engineering, we examined the new datasets to ensure the features were added successfully. The data types for newly added columns are correct, and the number of columns is also the number that we expected.

ETH missing values:		BTC missing values:	
date	0	date	0
open	0	open	0
high	0	high	0
low	0	low	0
close	0	close	0
volume	0	volume	0
quote_asset_volume	0	quote_asset_volume	0
number_of_trades	0	number_of_trades	0
taker_buy_base_asset_volume	0	taker_buy_base_asset_volume	0
taker_buy_quote_asset_volume	0	taker_buy_quote_asset_volume	0
daily_return	1	daily_return	1
log_return	1	log_return	1
volatility_7	7	volatility_7	7
volatility_14	14	volatility_14	14
volatility_30	30	volatility_30	30
MA_7	6	MA_7	6
MA_14	13	MA_14	13
MA_30	29	MA_30	29
EMA_7	0	EMA_7	0
EMA_14	0	EMA_14	0
EMA_30	0	EMA_30	0
momentum_7	7	momentum_7	7
momentum_14	14	momentum_14	14
momentum_30	30	momentum_30	30
volume_MA_7	6	volume_MA_7	6
volume_MA_14	13	volume_MA_14	13
volume_MA_30	29	volume_MA_30	29
RSI_7	7	RSI_7	7
RSI_14	14	RSI_14	14
RSI_30	30	RSI_30	30
dtype: int64		dtype: int64	

We checked for missing values again, and there are missing values found in previously added columns. Since the highest number of missing values in a column is 30 and all the missing values found in certain columns are at the first 1, 7, 14 or 30 rows, we decided to remove the first 30 rows of the data in both cryptocurrencies' datasets.

```
BTC shape before: (1826, 30)
BTC shape after: (1796, 30)

ETH shape before: (1826, 30)
ETH shape after: (1796, 30)
```

After dropping unnecessary rows, the final cleaned datasets retain 1796 rows per column without any missing values.

2.4 Feature Engineering

The original dataset only had 10 variables, which were not enough to support forecasting for all four target variables. After serious consideration, 20 additional variables were created through feature engineering and added to the dataset. While not all variables will be used in the final model, they increase the depth and richness of our datasets. That said, they are added to better support the prediction of price, return, volatility, and trading volume.

The engineered variables can be grouped into several categories. Firstly, we have return-based features that are daily return and log return, which measure day-to-day relative

price change. Secondly, we had added the rolling volatility features that measure over 7, 14, and 30 days to capture market instability. Moving on, we also created trend-related variables using simple moving averages (MA) and exponential moving averages (EMA) over 7, 14, and 30 days to represent the market trend. In addition, momentum variables over the same rolling windows were added to capture the strength and direction of recent price changes. Relative Strength Index (RSI) variables for 7, 14, and 30 days were also included to reflect overbought and oversold conditions. Finally, moving averages of trading volume for 7, 14 and 30 days were created to represent changes in market participation over time.

Overall, these engineered features extend the dataset beyond raw price and trading activity variables. With them, the trained models will have better performances in capturing trend, momentum, volatility, and volume behaviour in BTC and ETH markets.

2.5 Exploratory Data Analysis

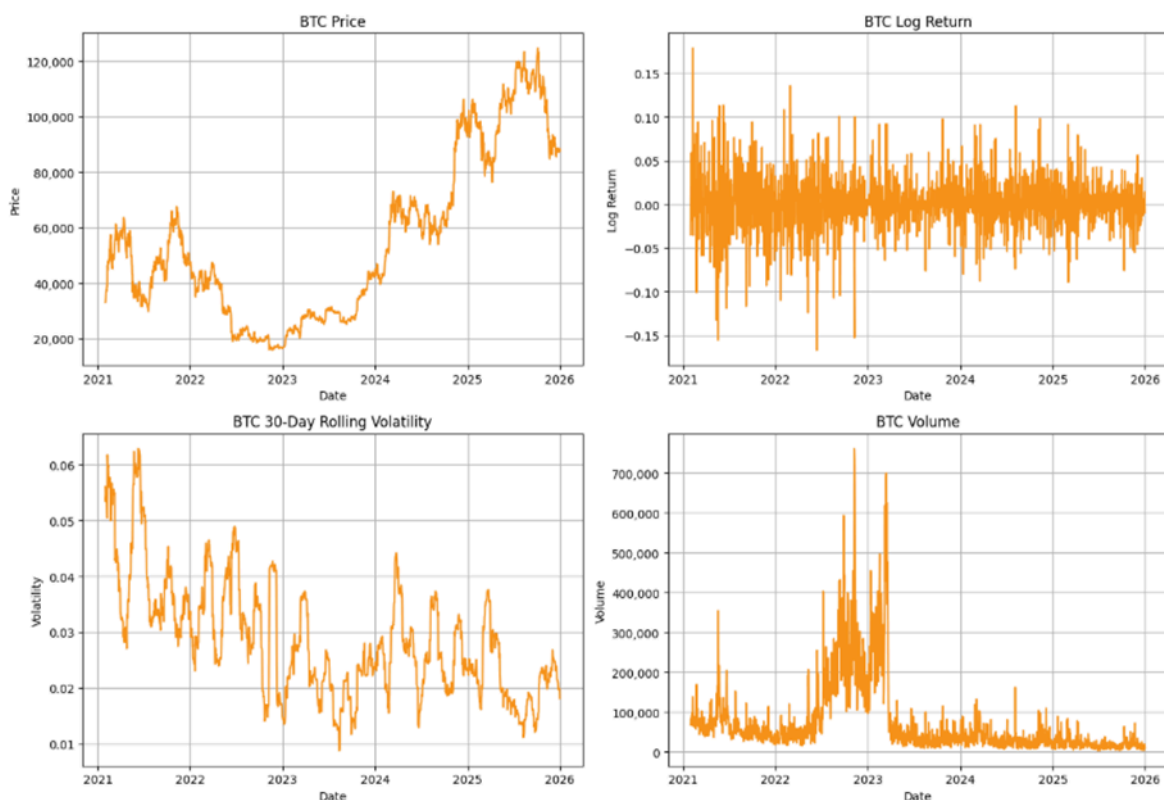


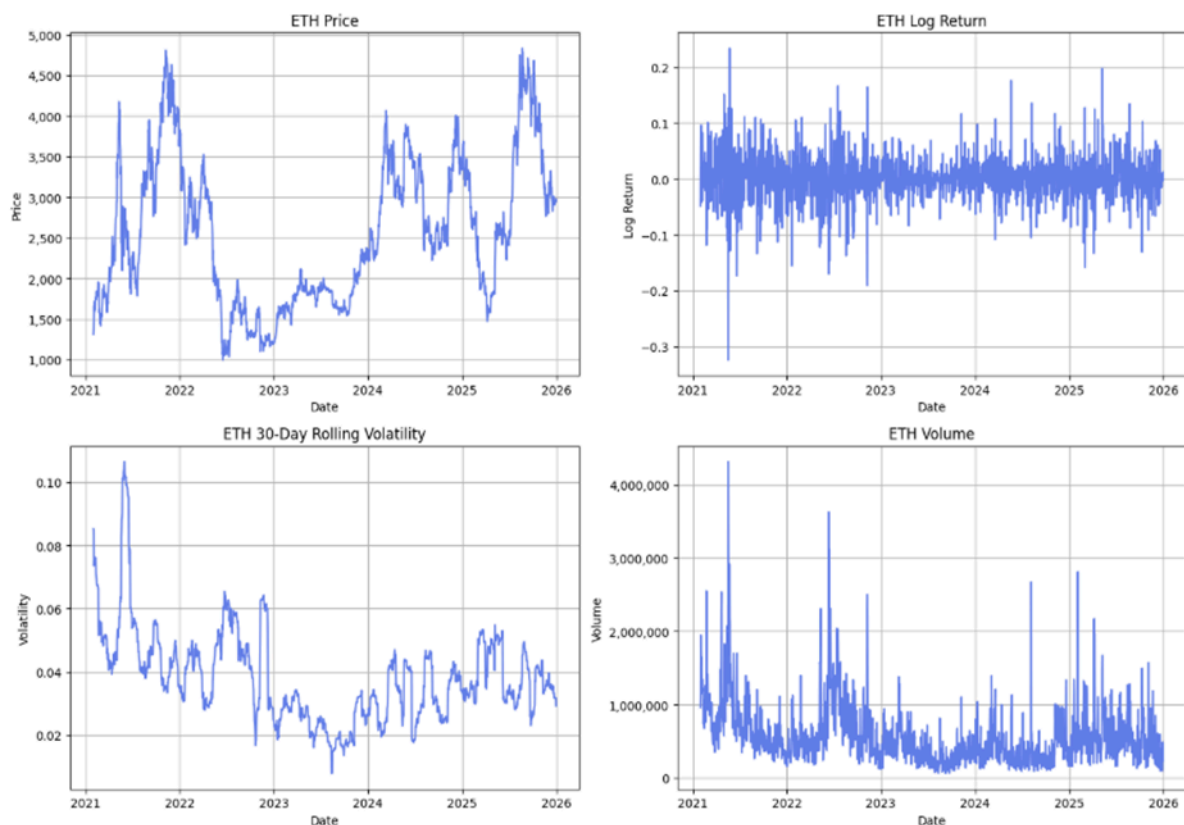
Figure above shows the main market behaviour of Bitcoin from 2021 to 2025 using four variables: price, log return, 30-day rolling volatility, and trading volume. The BTC price had experienced several strong rises and falls across the period, but in overall it remains an uptrend. This shows that Bitcoin has highly dynamic price behaviour and it is significantly influenced by changing market conditions over time.

Next, the log return plot is centred around zero, with many sudden spikes in both positive and negative directions. This means that although most daily changes are relatively small, there are still large movements that happen quite often.

Also, the 30-day rolling volatility plot shows that volatility is not constant. Instead, we need to expect that high-volatility periods tend to continue for some time.

Lastly, the trading volume plot shows several large spikes rather than a stable pattern. This indicates that market activity is uneven and often increases during major market events or strong price movements.

To conclude, these figures show that Bitcoin is a highly volatile asset with changing price, return, volatility, and trading activity over time.



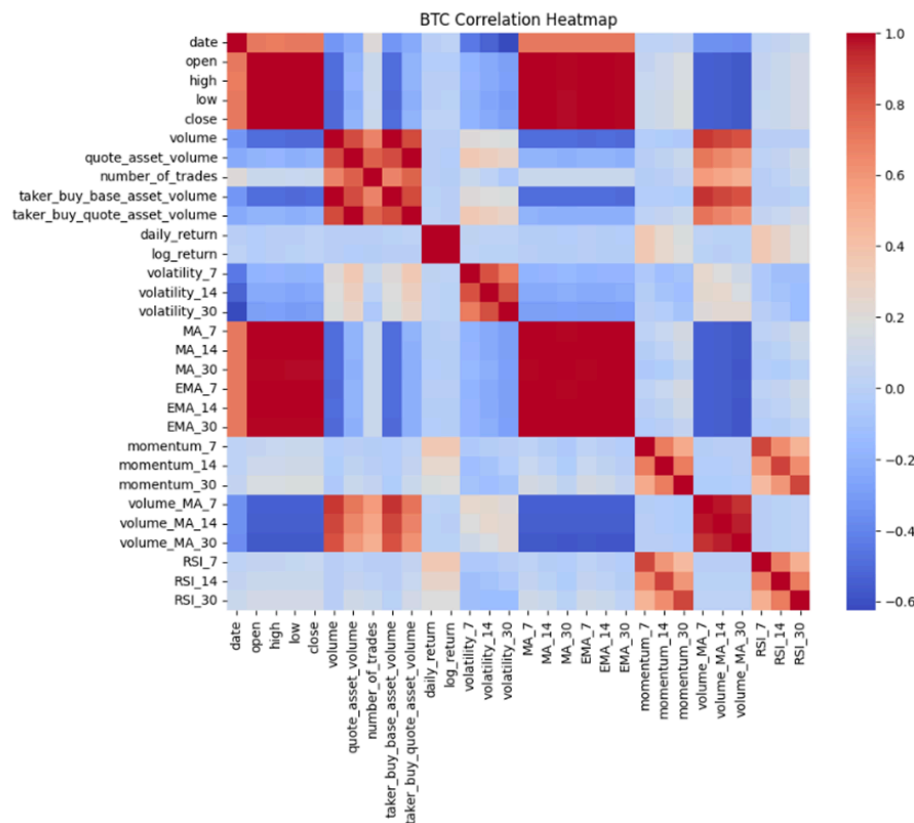
Unlike Bitcoin which experienced an overall upward trend over time, Ethereum's price appears to be more volatile. From the plot, it can be clearly observed that the price rises and falls rapidly within the 1500 and 4000-4500 price range. This indicates that there are stronger fluctuations in the Ethereum market.

Same as Bitcoin, the log return plot for Ethereum is also centred around zero, but the only difference is that it shows larger and more frequent extreme values. From here, we can say that Ethereum carries a higher level of short-term risk.

The 30-day rolling volatility plot supports that Ethereum is more volatile than Bitcoin, as it generally records higher volatility levels and stronger volatility clustering across the period.

Moreover, Ethereum's trading volume also shows frequent large spikes. What we know from the visualization is that the trading activity changes sharply over time instead of remaining stable.

Compared to Bitcoin, Ethereum seems to have more irregular market behaviour and higher overall risk level. Overall, these visualizations suggest that Ethereum is more volatile and reactive than Bitcoin.



From the Bitcoin correlation heatmap, it can be observed that the price-related variables, including open, high, low, and close are highly positively correlated with each other. This is expected since they all represent similar aspects of price movement. Plus, the moving averages and exponential moving averages also show strong positive correlations with the price variables. This is not surprising as they are highly dependent on price.

Next, the volume-related variables, such as trading volume, quote asset volume, and number of trades, have shown moderate to strong positive correlations among themselves. What we can tell is that trading activity variables tend to move together, no matter the market participation.

On the other hand, we can also find that variables such as log return, momentum, RSI, and volatility show weaker or more mixed correlations with price variables. While this observation makes sense, as these features capture different aspects of the market rather than simply following the price levels.

To conclude, the Bitcoin heatmap shows the correlations between variables, and those engineered features have provided additional information which may be useful for improving model performance.

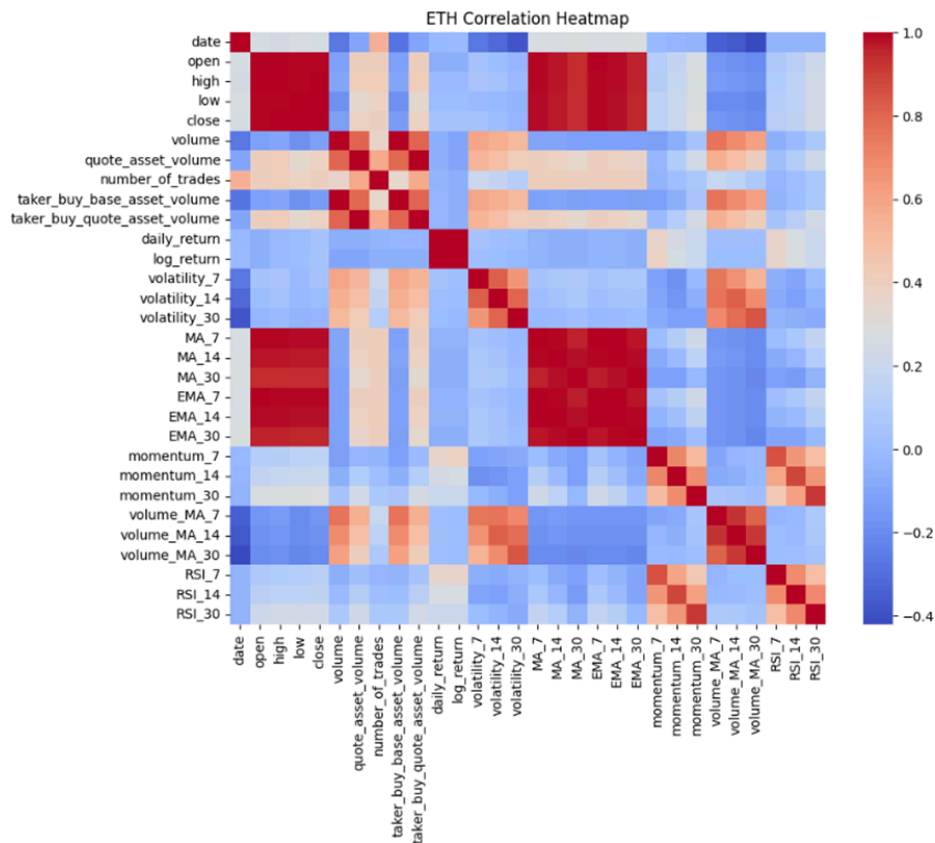


Figure above presents the correlation heatmap for Ethereum. Similar to Bitcoin, it can be observed that there are strong positive correlations among price-related variables. MA and EMA also show high correlations with these price variables as they are effectively capturing the overall price trend.

Again, the volume-related variables show moderate to strong correlations with each other, and we can indicate that trading activity tends to increase or decrease together in the Ethereum market.

Compared to Bitcoin, Ethereum shows slightly more variation in the relationships between some engineered features like volatility, momentum, and RSI. These variables are less strongly correlated with price because they capture different market dynamics such as risk and market sentiment.

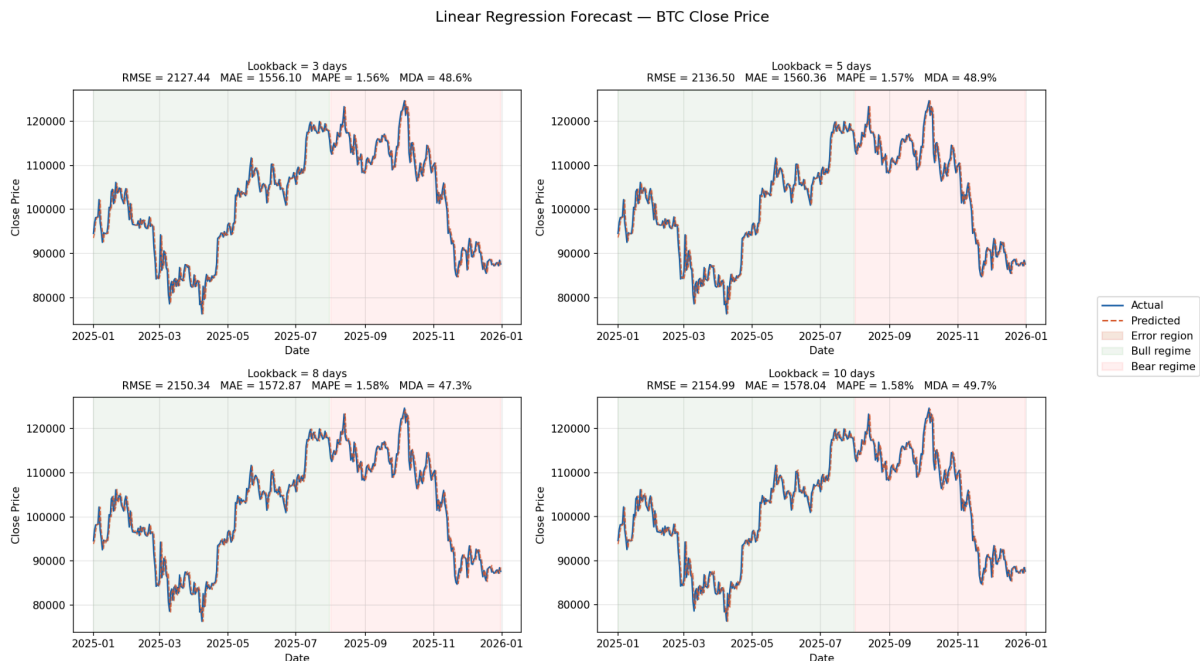
Overall, the heatmap suggests that Ethereum shares a similar correlation structure with Bitcoin but with slightly more variability. What we can conclude here is that Ethereum has a more volatile and dynamic market behaviour.

3.0 Initial Findings

So far, we have implemented LSTM, linear regression, and random forest regression models. These initial results were used to test the quality of the collected data and evaluate the baseline performance of each model across the selected targets. At this stage, the models are not fully optimised, but rather provide an early understanding of predictability within the dataset. Further experimentation, tuning, and refinement will be carried out in later stages to improve performance and better capture underlying market dynamics.

3.1 Price Behaviour

Linear regression



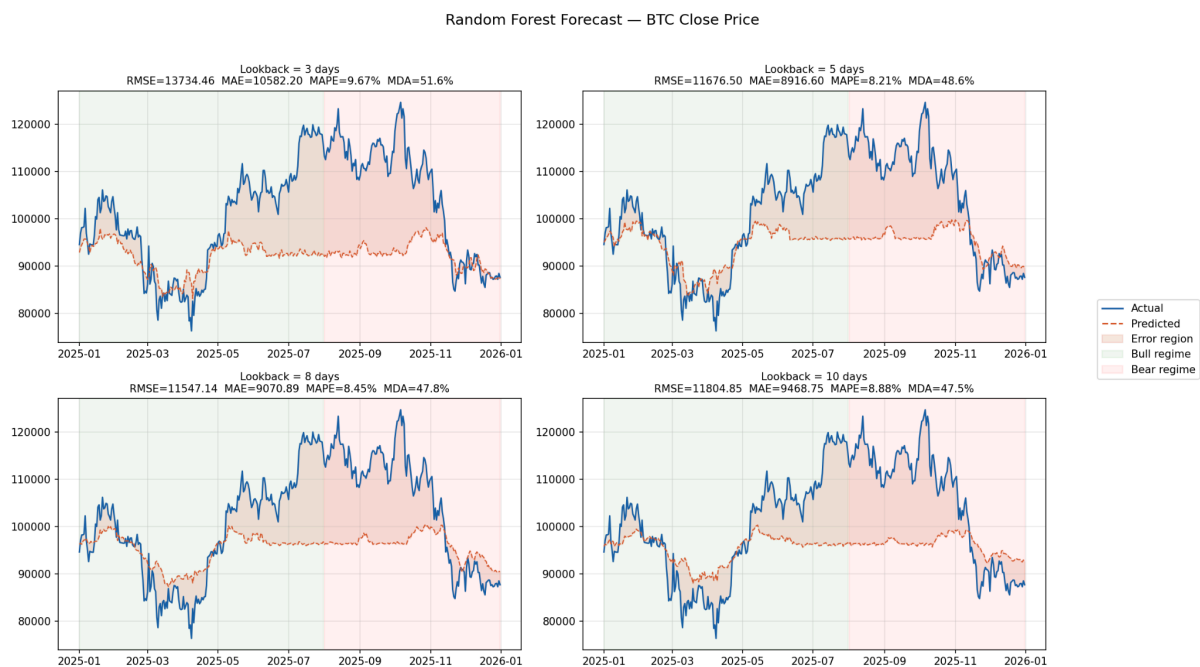
The baseline Linear Regression model was used to predict BTC close price using close price, log return, trading volume, and 7-day volatility with lookbacks of 3, 5, 8, and 10 days. As with volatility, the grouped plots make the predictions appear almost 1:1 with the actual series, since the predicted and actual lines overlap very closely across most of the test period.

However, this visual similarity should be interpreted carefully. Close price is a highly persistent series, so the model was able to track the general price level and broader trend well, which makes the graph look stronger than the model's true forecasting ability. This is reflected in the very low MAPE values (around 1.56%–1.58%), but also in the MDA remaining close to 50%, showing that the model was much better at following the overall path of price than at consistently predicting day-to-day directional changes.

Among the four settings, the 3-day lookback performed best on RMSE (2127.44), MAE (1556.10), and MAPE (1.56%), while the 10-day lookback produced the highest MDA (49.73%). Overall, these results suggest that Linear Regression was effective at tracking the

general level of BTC close price, but that its predictive strength was more limited than the graph alone may initially suggest.

Random forest regression



The random forest model was used to predict BTC close price using close price, log return, trading volume, and 30-day volatility as input features, with lagged versions across lookbacks of 3, 5, 8, and 10 days. The same tuned hyperparameters were applied to maintain consistency across experiments.

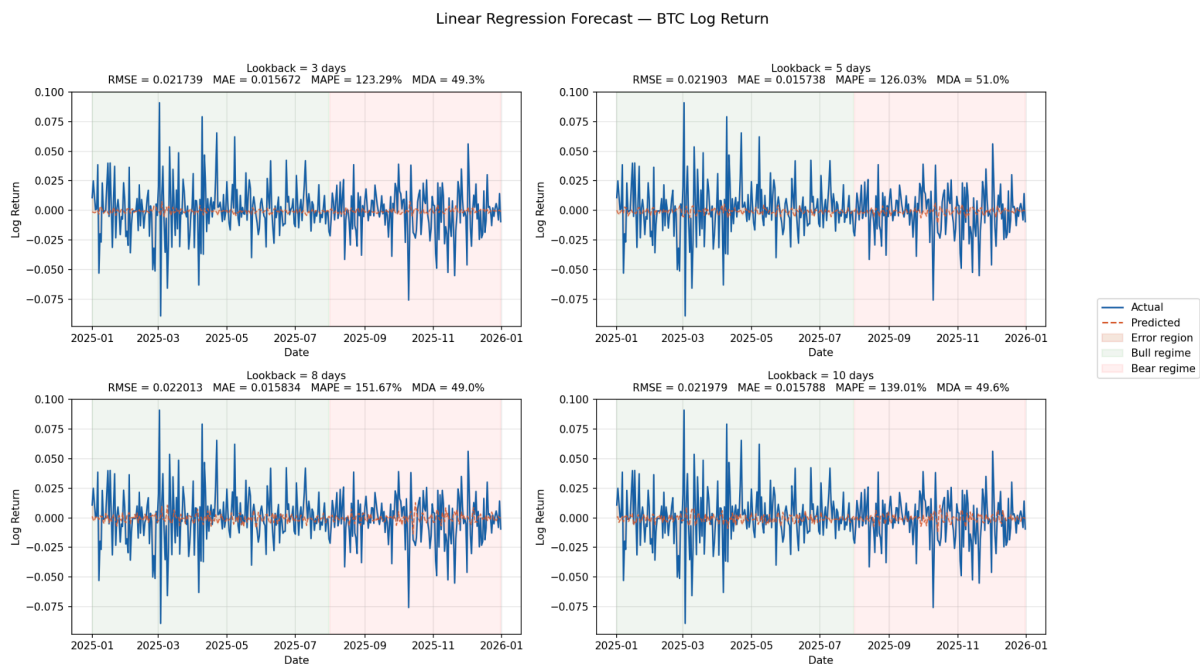
Compared to volatility, price proved to be a more challenging target due to its non-stationary nature and large fluctuations over time, ranging roughly between \$80,000 and \$125,000 during the test period. The model achieved moderate performance, with MAPE values between approximately 8% and 10% and relatively high RMSE across all lookbacks. While the model was able to capture the general trend of price movements, the predicted series appeared noticeably smoother than the actual values, indicating difficulty in modelling short-term fluctuations and sharp changes.

The 8-day lookback produced the lowest error, suggesting that a slightly longer short-term memory helps capture recent momentum. However, directional accuracy (MDA) decreased as the lookback increased, with the highest value at the 3-day lookback. This suggests that shorter lookbacks are more effective for capturing immediate directional changes, while longer lookbacks introduce lag and reduce responsiveness.

Overall, the findings indicate that the random forest model can approximate broad price trends but struggles with short-term dynamics and sudden market movements, reinforcing that price is a more complex and less predictable target than volatility.

3.2 Return

Linear regression

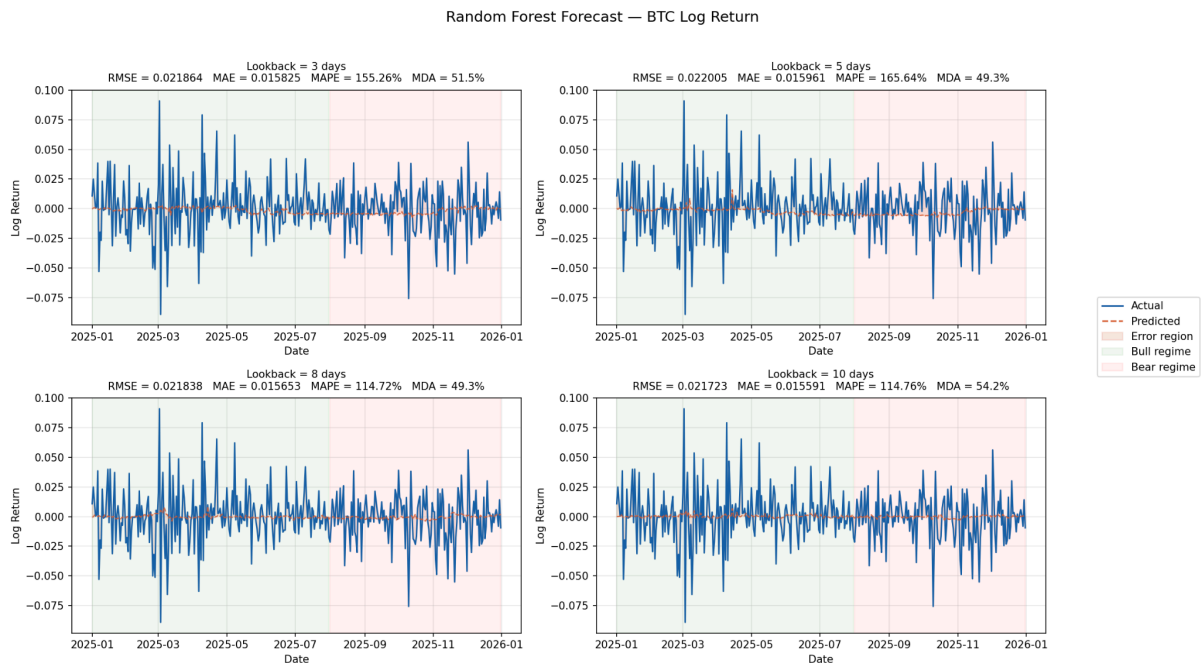


The baseline Linear Regression model was used to predict BTC log return using close price, log return, trading volume, and 30-day volatility with lookbacks of 3, 5, 8, and 10 days.

As shown in both the metrics and the grouped graph, BTC log return was difficult to predict. RMSE and MAE were very similar across all lookbacks, showing that changing the historical window had little effect on performance. MAPE was very high in all cases, which is expected for return data because values are often close to zero, making percentage error unstable.

The predicted series stayed much flatter and closer to zero than the actual series, meaning the model could follow the general average level of returns but failed to capture the larger positive and negative spikes. Overall, the 3-day lookback gave the lowest RMSE and MAE, while the 5-day lookback had the highest MDA (50.96%). However, the differences were very small, and performance remained weak overall, indicating that Linear Regression has limited predictive power for BTC log return.

Random forest regression



The random forest model was used to predict BTC log return using close price, log return, trading volume, and 30-day volatility as input features, with lagged versions across lookbacks of 3, 5, 8, and 10 days. The same tuned hyperparameters were used for consistency across all experiments.

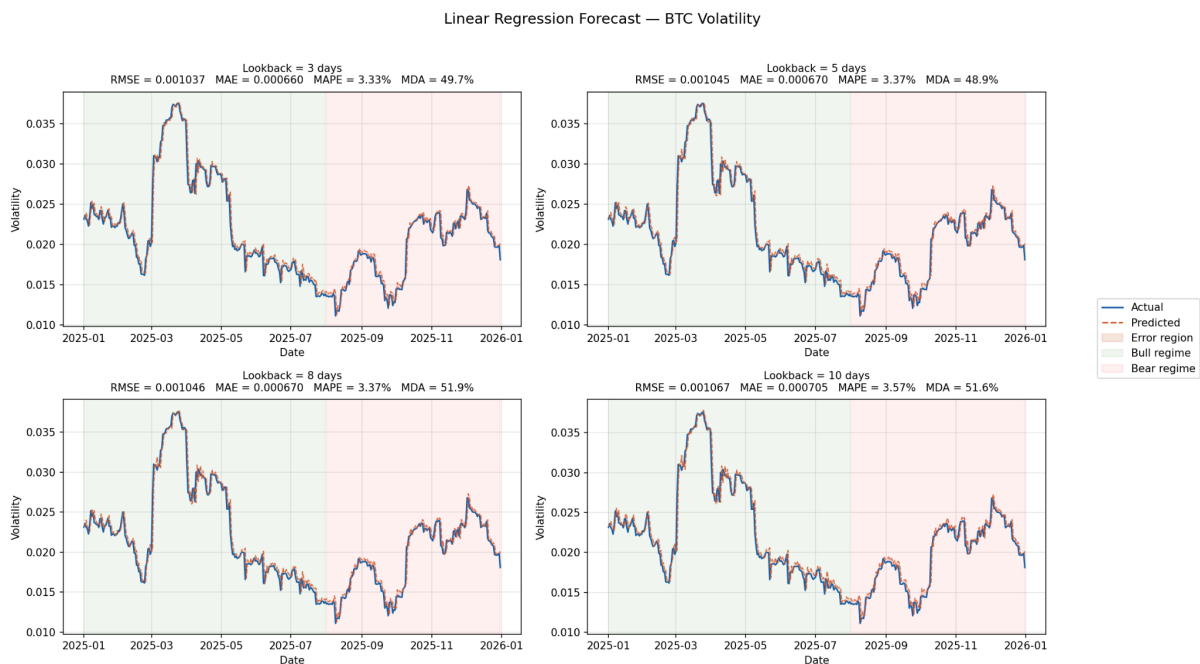
Compared to volatility and price, returns were significantly more difficult to predict. RMSE and MAE were very similar across all lookbacks, indicating that changing the historical window had limited impact on performance. MAPE was extremely high in all cases, reflecting the instability of percentage-based error for return series where values are close to zero, making RMSE, MAE, and MDA more meaningful evaluation metrics.

The predicted series remained much flatter than the actual log returns, showing that the model struggled to capture the large day-to-day fluctuations. While it could approximate the general level of returns around zero, it failed to reproduce the magnitude of sharp movements, which are often driven by external market factors not captured in the features.

The 10-day lookback performed slightly better overall, achieving the lowest RMSE and MAE and the highest MDA ($\approx 54\%$). However, improvements were modest, and overall performance remained weak. These results highlight that random forest has limited predictive power for BTC log returns, confirming that returns are one of the most difficult targets to model reliably.

3.3 Volatility & Volume Patterns

Linear regression

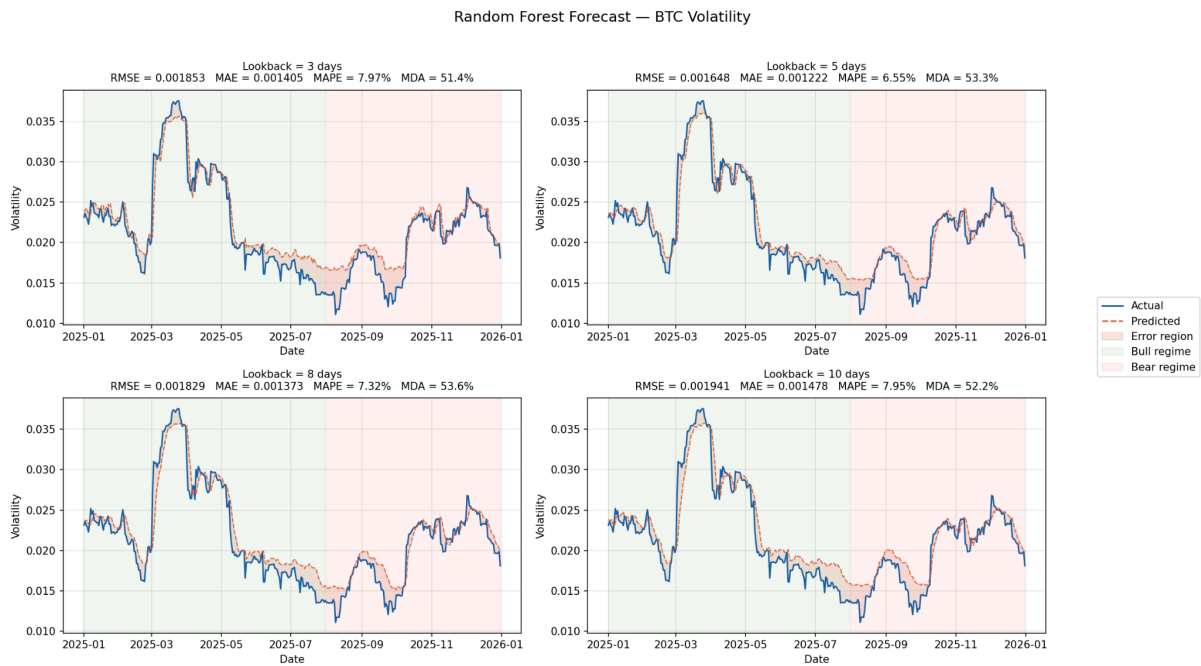


The baseline Linear Regression model was used to predict BTC volatility using close price, log return, trading volume, and 30-day volatility with lookbacks of 3, 5, 8, and 10 days. At first glance, the grouped plots make the predictions appear almost 1:1 with the actual series, as the two lines overlap very closely across most of the test period.

However, this visual overlap should be interpreted carefully. Because volatility moves within a relatively narrow range and the full-year graph compresses small differences, the plot can make the model look stronger than it really is. In practice, the model was better at tracking the general level and shape of volatility than at consistently predicting day-to-day directional changes, which is reflected in the MDA values staying close to 50%.

Even so, volatility was still one of the stronger targets for the baseline model. The 3-day lookback gave the lowest RMSE (0.001037), MAE (0.000660), and MAPE (3.33%), while the 8-day lookback achieved the highest MDA (51.92%). Overall, these results suggest that Linear Regression was effective at following the broader volatility pattern, but its predictive strength was more moderate than the graph alone may suggest.

Random forest regression



The random forest model was used to predict 30-day realised volatility for BTC using a set of core features consisting of close price, log return, trading volume, and historical volatility. Lagged versions of these features were constructed across four lookback windows (3, 5, 8, and 10 days) to capture temporal dependencies in the data. The model was configured using tuned hyperparameters (100 estimators, maximum depth of 5, and log2 feature selection), providing a balance between model complexity and generalisation.

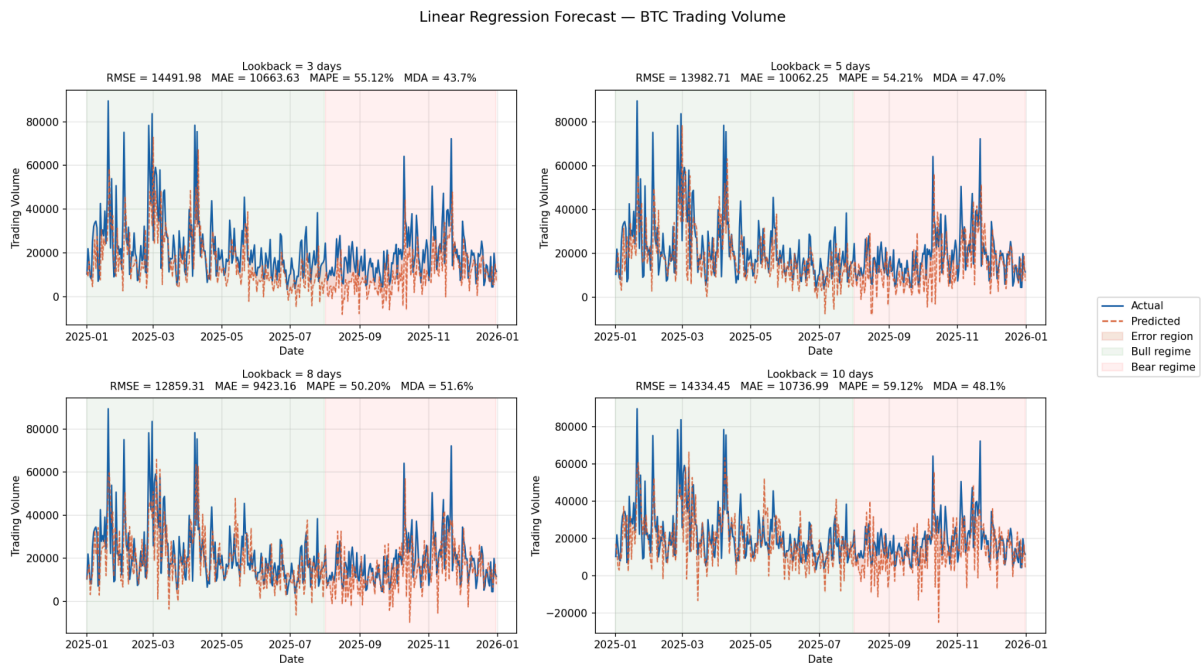
The results indicate that volatility is the most predictable variable among all targets explored. The model achieved consistently low error values, with MAPE ranging approximately between 6% and 8%, and RMSE remaining stable across all lookback configurations. The predicted series closely follows the actual volatility, particularly during sustained periods of high or low volatility, demonstrating the model's ability to capture the overall level and trend.

This strong performance can be explained by the nature of 30-day volatility, which is a smoother and more stable series compared to shorter-term measures. The persistence of volatility over time, commonly referred to as volatility clustering, provides a strong signal for the model, allowing it to learn patterns effectively even with a relatively simple feature set.

Performance differences across lookbacks were minimal, with the 5-day and 8-day windows producing slightly better results. This suggests that 30-day volatility relies more on recent behaviour rather than requiring long historical context. However, directional accuracy (MDA) remained moderate at around 51–54%, indicating that while the model is effective at estimating the magnitude of volatility, it struggles to consistently predict short-term increases or decreases.

Overall, the findings show that the random forest model performs well when applied to smoother, long-term volatility measures. The stability and persistence of the target variable make it easier to model, resulting in strong predictive performance even without the use of complex architectures or extensive feature engineering.

Linear regression

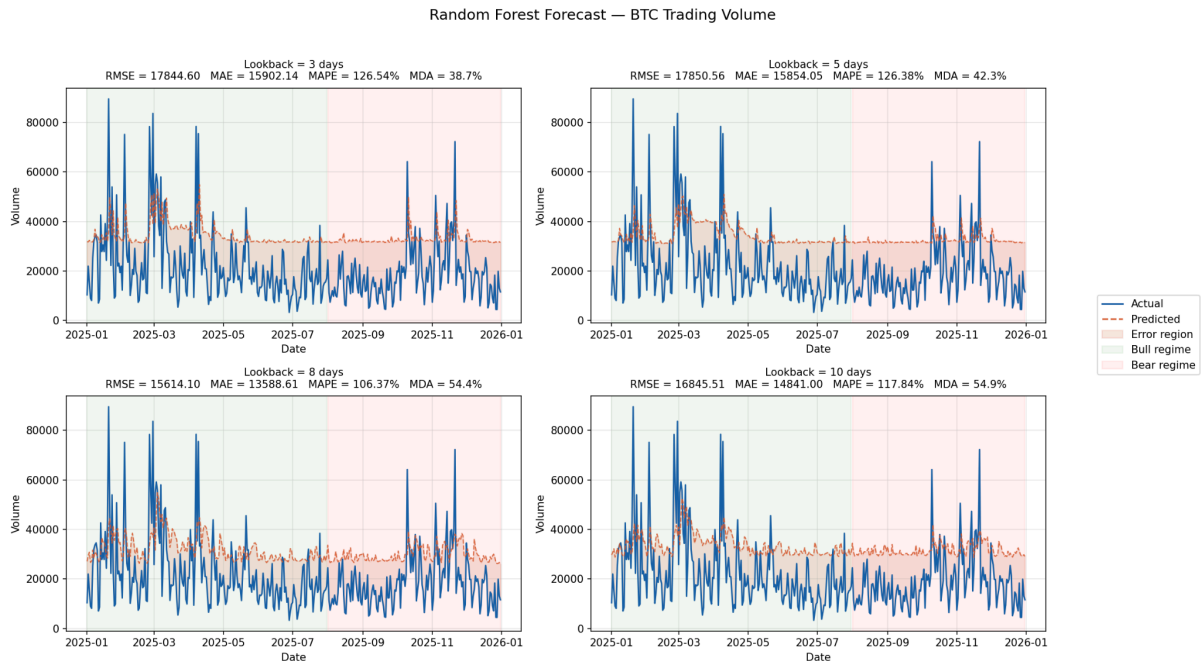


The baseline Linear Regression model was used to predict BTC trading volume using close price, log return, trading volume, and 30-day volatility with lookbacks of 3, 5, 8, and 10 days. Compared with close price and volatility, trading volume was more difficult to model, although the results were more stable and realistic than those for log return.

From both the graph and the evaluation metrics, the model was able to follow the general movement and timing of changes in trading volume, but it still struggled to capture the full magnitude of large spikes, especially during periods of unusually high activity. This suggests that the model learned the broader pattern of volume behaviour, but not the more extreme short-term fluctuations.

Among all configurations, the 8-day lookback performed best overall, producing the lowest RMSE (12859.31), lowest MAE (9423.16), lowest MAPE (50.20%), and the highest MDA (51.65%). Even so, the error values remained fairly high, showing that BTC trading volume is still a challenging target for a simple linear model and that its predictive performance remains limited overall.

Random forest regression



The random forest model was used to predict BTC trading volume using close price, log return, trading volume, and 30-day volatility as input features, with lagged versions across lookbacks of 3, 5, 8, and 10 days. The same tuned hyperparameters were used for consistency across all experiments.

Among all targets, trading volume was one of the most difficult to model. Error values remained high across all lookbacks, with MAPE staying above 100% in every configuration, showing that the model struggled to estimate the magnitude of daily volume accurately. The predicted series was much smoother than the actual data, indicating that the model mainly captured average volume levels rather than the sharp spikes and sudden drops visible in the market.

Performance improved at the longer lookbacks, particularly 8 and 10 days. The 8-day lookback produced the lowest RMSE, MAE, and MAPE, while the 10-day lookback achieved the highest MDA at approximately 55%. This suggests that slightly longer short-term memory helps the model capture broader trading activity patterns, even though the improvement remains limited.

Overall, the findings show that random forest has weak predictive power for BTC trading volume. While longer lookbacks provide some improvement, the model still struggles with the highly irregular and spike-driven nature of volume, making it a much harder target to predict than volatility and price.

4.0 Difficulties & Risks

4.1 Difficulties

The experimental phase revealed several key difficulties when applying machine learning models to cryptocurrency data.

One of the main difficulties was modelling variables with very different statistical behaviour. From the results, volatility was relatively smooth and predictable, while return and trading volume were highly irregular. For example, return predictions across both linear regression and random forest remained close to zero and failed to capture large spikes, indicating that the models were unable to learn meaningful patterns from this variable. In contrast, volatility showed more stable behavior and was easier for the models to follow.

Another major difficulty was the limited ability of models to capture short-term fluctuations. For price prediction, both linear regression and random forest were able to follow the overall trend but produced smoother outputs than the actual data. This means that while general movement was captured, the models struggled with sudden changes and sharp market movements.

Model behaviour also varied significantly depending on configuration. Different lookback windows produced noticeably different results, and no single configuration performed best across all targets. This required repeated testing and adjustment to find suitable settings for each variable.

In addition, instability was observed in more complex models. Some configurations, particularly more aggressive LSTM setups, produced unrealistic or extreme predictions. This suggests that the models were highly sensitive to parameter choices and, in some cases, were unable to learn stable patterns from the data.

Finally, it was consistently observed that models were better at tracking overall trends than predicting direction, with MDA values remaining close to 50% across most experiments. This highlights a limitation in the model's ability to provide actionable predictions.

4.2 Risks

Data Risks

A key data-related risk identified from the results is the lack of strong predictive structure in certain variables. Return and trading volume showed behaviour similar to random noise, with frequent spikes and no consistent patterns. This increases the risk that models will not be able to learn meaningful relationships, leading to weak predictive performance.

Another data risk is the absence of external influencing factors. The data set only includes historical market variables, while real cryptocurrency markets are heavily affected by external events such as news, regulation, and market sentiment. This was evident in the results, where models failed to capture sudden spikes and sharp movements.

Modelling Risks

From a modelling perspective, one major risk is overfitting, particularly when using more complex models. This was observed when certain configurations produced extreme or unrealistic predictions, suggesting that the model was fitting noise rather than general patterns.

There is also a risk of underfitting, where models produce overly smooth or flat predictions, this was clearly seen in return and volume predictions, where the models approximated average values rather than capturing actual variation in the data.

Another important risk is misleading performance evaluation. For example, models achieved low error values in some cases by closely following the general trend, but still failed to predict direction accurately, as shown by MDA values close to 50%. This means that relying only on error metrics may overstate model performance.

Finally, there is a broader risk that models developed on historical data will not generalise well, as cryptocurrency markets are highly volatile and subject to sudden changes that cannot be captured by past data alone.

4.3 Mitigation

Several strategies were applied to address these difficulties and reduce overall risk.

First, multiple models and configurations were tested across each target variable rather than relying on a single approach. This was necessary because the results showed clear differences in predictability across variables, requiring the modelling approach to be adapted accordingly.

Second, Multiple evaluation metrics, including RMSE, MAE, MAPE, and MDA, were used to assess performance from different perspectives. This helped identify cases where models appeared accurate in magnitude but failed to predict direction.

Different lookback windows were also tested to determine the most suitable temporal structure for each variable. As performance varied across configurations, this process helped reduce sensitivity to parameter choices and improve consistency.

In addition, model outputs were interpreted carefully rather than relying only on numerical metrics. For example, flat predictions for return and smoother predictions for volume were recognised as limitations of the data rather than indicators of strong model performance.

Finally, the limitations of the data set were explicitly acknowledged, particularly the absence of external influencing factors. This ensures that conclusions remain realistic and aligned with the observed behaviour of the models.

5.0 Updated Project Plan

5.1 Progress so far

The team has accomplished considerable advancements since the time of submitting the original project proposal. The first two phases have been successfully completed by collecting historical cryptocurrency data from Binance. After the cleaning and preparation of the dataset, a comprehensive feature set was created to allow modelling. This includes the typical return based features; volatility metric moving averages Exponential Moving Averages; Momentum Indicators; RSI; and volume-based features. Collectively this group of features will assist in identifying how the cryptocurrency markets behave.

In addition to creating the features, the team has completed an Exploratory Data Analysis (EDA). The EDA provided the team with a good understanding of the internal structure of the data collected. It can be noted that several key properties of the data were found in both Bitcoin and Ethereum datasets. Additionally, the correlation analysis showed there exists a relationship between some of the variables collected and validated that engineered features do provide additional useful knowledge in addition to raw market data.

To date, the team has used a number of modeling methods including Linear Regression, Random Forest Regressions, and Long Short Term Memory Models. All of these models have been tested against each of the four target variables. In evaluating performance under different conditions a variety of configurations have been evaluated utilizing different look back windows. To date, it appears that volatility is the easiest to predict among the three target variables. However, predicting return and trading volume appear to be much harder than predicting volatility. As for predicting price, the results indicate moderate success. While it is possible to capture overall trends in the price of cryptocurrencies, predicting specific short term prices remains difficult. Overall, the project has made significant progress towards developing a solid foundation for future model development and analysis.

5.2 Changes

Based on the work done thus far, a number of changes have been made to the original project plan.

It took longer than anticipated to do data preprocessing and feature engineering due to the complexity of working with time-series data. Therefore, the modeling process was slightly delayed as a result.

The modeling strategy has been modified. Rather than apply one model configuration to all variables, there will be separate model configurations applied for each target variable. This is needed as each target variable exhibits different characteristics in terms of level of predictability and/or type of behaviour and relationships.

As an example; Volatility benefits from using a less aggressive model configuration than does Price Prediction which benefits from a more aggressive model configuration (one that

captures both long term trends and also short term movements). This illustrates why models should be constructed based on how well they match the behavior of the data being analyzed.

Lastly, a greater amount of effort has been focused on evaluating multiple configurations/models instead of focusing on a singular approach.

5.3 Next steps

The next stages in this project will involve refinement of the model's performance and completion of any outstanding analysis. Additional experiments will be carried out for further optimization of the model configuration and improvement of the predictive ability of the model with respect to all target variables. Additionally, comparative analyses will be undertaken based on the results from the various models developed to date so as to determine which is the most successful approach to modelling each variable.

Further assessment will take place of the models utilizing a variety of measures of performance to provide an enhanced understanding of their efficiency. Following this stage, the project will proceed toward producing the final report and presentations; therefore, findings will be presented effectively and appropriately illustrated through relevant graphs and tables etc. and analysis. At present, the project appears to be progressing in accordance with its planned deadlines.